Compromising protected domain environments

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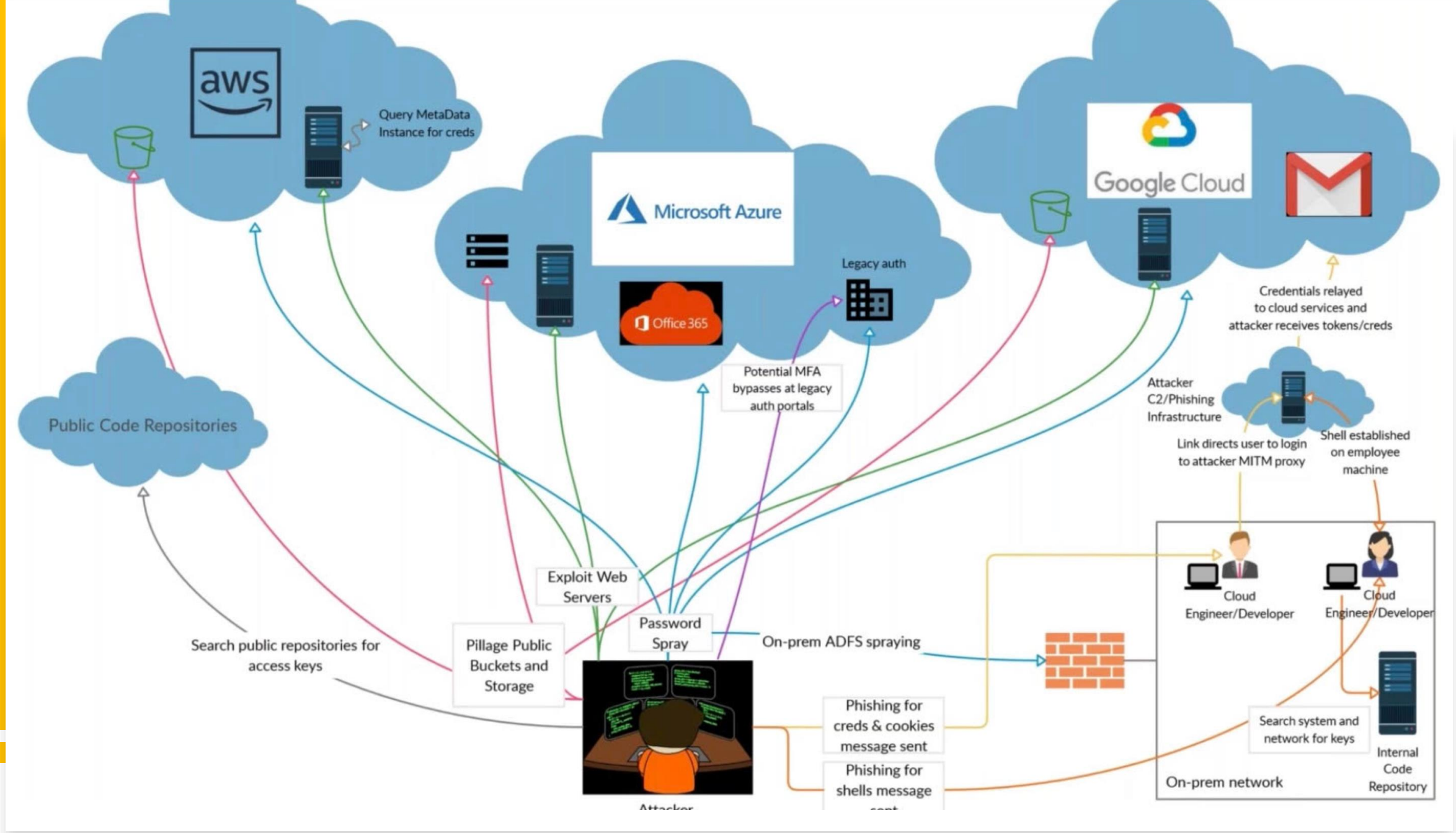
Typical attack scenario

- Reconnaissance of external facing domains, services, applications and leaked credentials with an intent to find the entry point.
- Delivery of crafted phishing emails with malicious attachments and links to the employees' mail addresses.
- Credentials reuse or execution of the payloads and establishing Command&Control (C2) communication.
- Situational analysis on the hosts with further domain enumeration.
- Lateral movement.
- ….
- Actions on the target (data exfil, DoS, ransomware).



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Taken from BHIS talk

Agenda

- Payload delivery and execution.
- On-disk and In-memory detection evasion.
- Active Directory and Kerberos.
- Lateral movement techniques and traces.
- Constrained and Unconstrained delegation.
- GPO abuse.
- Kerberos on Linux.

Attackers' mind-list

- Mail server config, filtering, malware detection, blacklists
- Host based protection and detection: FW, AV/ AMSI/EDR, App/ Device Whitelisting, Group Policies
- Network protection, segregation and anomaly detection.
- Threat detection and response: SIEM / SOAR.
- IAM, security ~~mis~~configurations, patching and more.

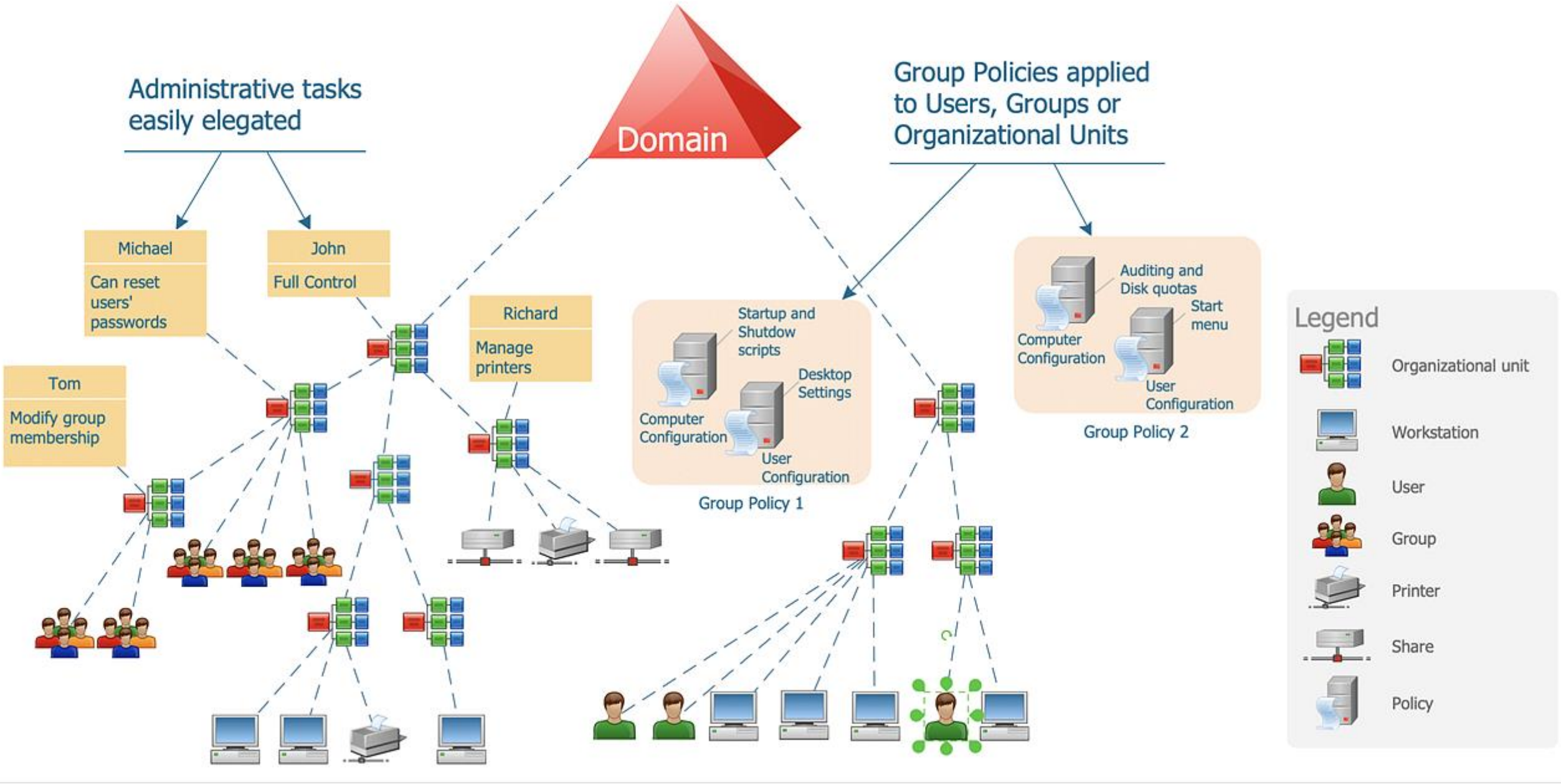
AV / EDR Evasion

- On-Disk:
  - Payloads tweaking, packing, obfuscation or encryption
  - Injection into legit files (PE infection)
  - Shellcode invokers
- In-Memory:
  - Process injection
  - Reflective loading
  - Process hollowing
  - Inline hooking
  - Parent-Process Relationships (PPID-PID)
  - Injection method
  - Thread start address (associated with file)
  - Memory permissions (no RWX)
  - Memory content (avoid known strings and patterns)

- Disable AV or exempt path to the payload.
- Use custom tools.

Demo. AV bypass.

- PowerShell tweaks
- Payload encryption
- Shellcode runner



Source

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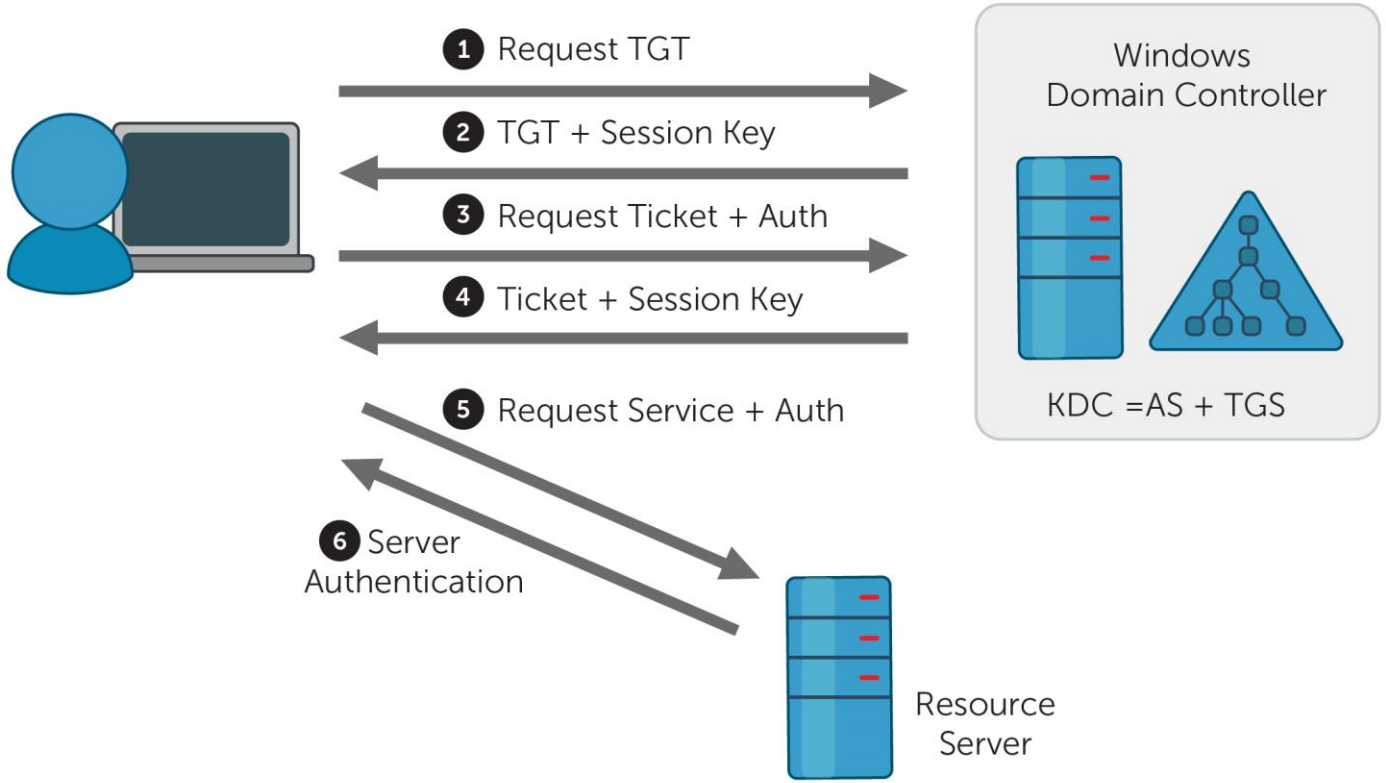
Kerberos from an attacker’s perspective.



1 - Overpass the hash, AS-REP roasting

2 - Golden ticket

3 - Kerberoasting

4 - Silver Ticket

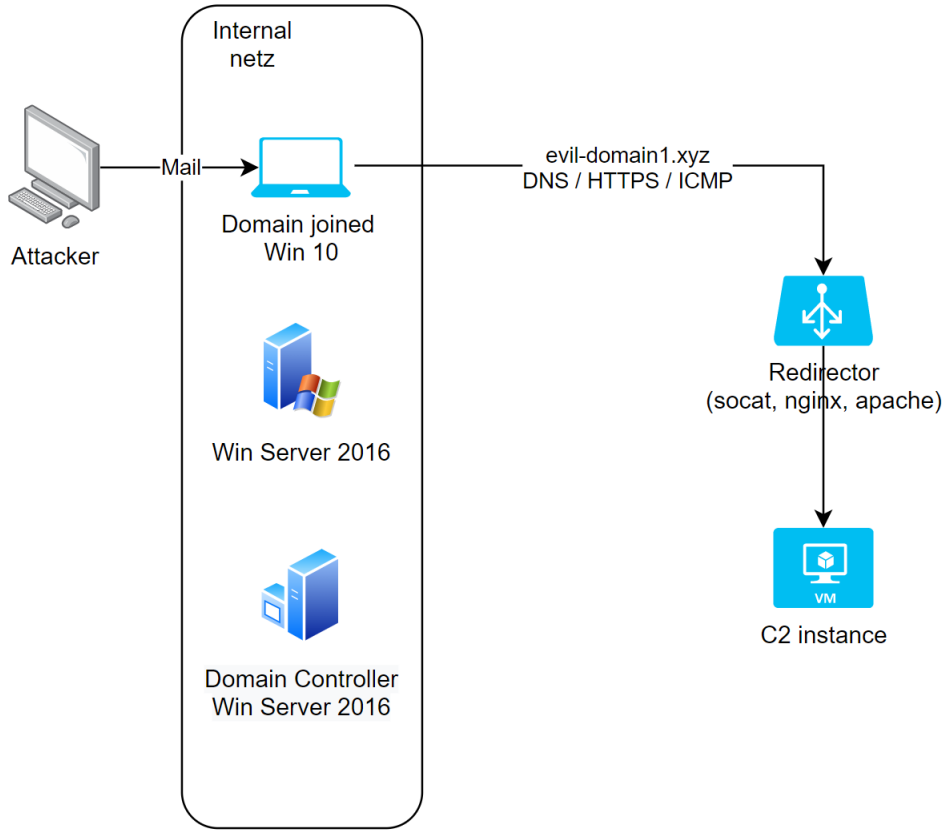
5 - Pass the ticket

Lateral Movement

Techniques	Protocol / Default Ports
PsExec, Service Control	SMB / 445
WMI, Scheduled Tasks	RPC / 135
Remote Desktop	RDP / 3389
Windows Remote Management	WinRM / 5985, 5986
DCOM	RPC / 135, 1024++
Admin / Dev tools, code repos Configuration / Orchestration tools WSUS, SCCM, AV, GPO Vulnerable Apps and Infras DB links …	

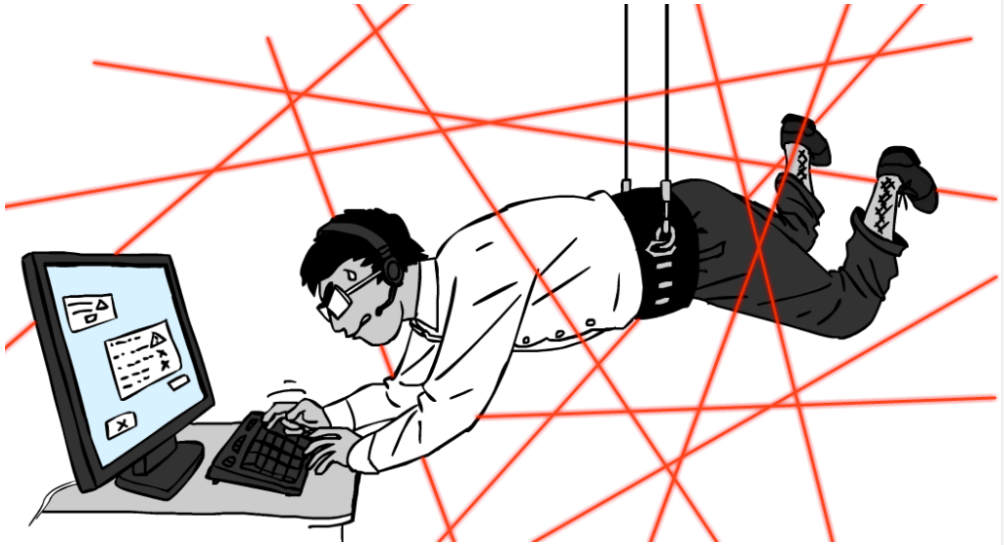
Demo. Lateral Movement.

- Service creation and hiding
- WMI + regsvr32
- MsBuild
- UAC bypass
- Unconstrained delegation + Printer Bug = DA
- Constrained delegation
- GPO Abuse

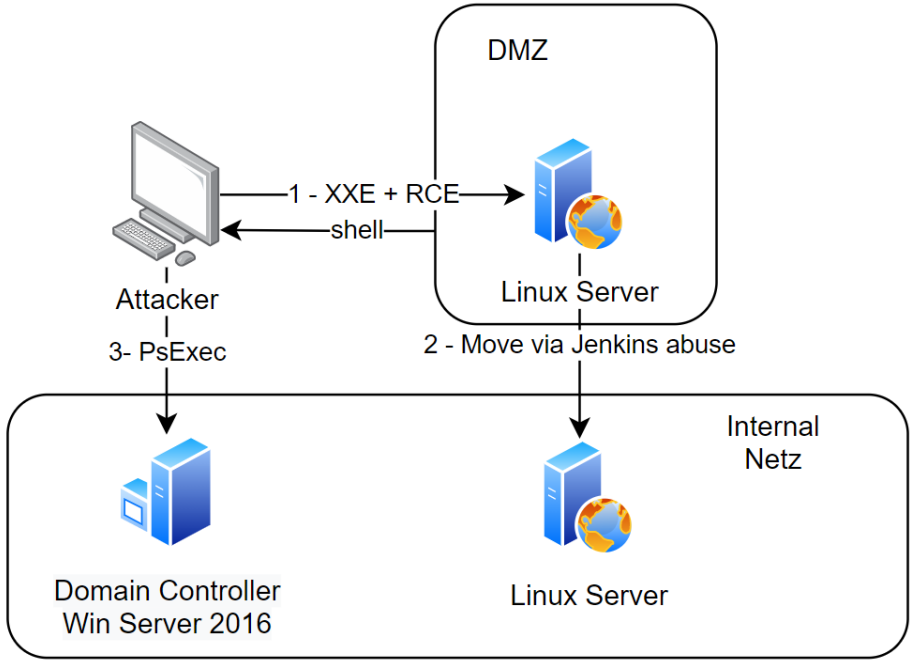


Detection Bypass During Post Exploitation

- Use of custom tools and techniques.
- Avoid using powershell.exe
- Avoid querying endpoints
- Blend into normal traffic and behavior.
- LoL bins and scripts.
- Use of less monitored systems (Unix, IoT).
- Speed (Not Petya, Infection Monkey).
- Disable agent / logging.



Demo. From Linux to Domain Admin.



- XXE
- LFI + Command Execution
- Linux Privilege Escalation
- Port Forwarding techniques
- Jenkins exploitation
- Abuse Kerberos on Linux
- Pass the ticket with PSExec

Summary.

- Most of known malicious activity can be detected for the price of events to review.
- Most of known detection techniques can be avoided.
- Domains are complex and full of abuse possibilities if not analyzed regularly for security issues.
- Clouds make the attackers' life more interesting.